



MK27-2, Amazon Prime Air's latest delivery drone, with a unique hexagonal design for better stability and six degrees of freedom (Courtesy: Amazon)

remotely by trained personnel who can take control at any time, if necessary, and also speak to the users.

What's in the bots

Autonomous last-mile delivery bots are very similar to self-driving cars. They require almost all the technology, plus some more. Each bot contains a variety of sensing technologies, such as state-of-the-art thermal cameras, point cloud sensors, lidar and radar, inertial ultrasonic sensors, and more. Then, there are processing units, which include central processing units, graphics processing units, neural network accelerators, field-programmable gate arrays, and embedded integrated circuits.

That said, software forms the backbone of any autonomous system. "We have a massive software system of millions of lines of code, both in the robot and in the cloud. In simple terms, each robot has an awareness system. It can see what's going on, avoid objects, detect objects, brake, accelerate, turn corners. It identifies where it is and signals to central locations if there's a problem with the device. And the cloud software handles things like payments, route planning, fleet orchestration. It monitors the robot health and energy charge and ensures that the correct number of robots arrive

at the right time at the right locations. It's a choreography of things in the cloud as well as things on the ground," explains Alastair Westgarth, CEO of Starship Technologies, in a Forbes interview.

The robots almost always have a data connection, either cellular or Wi-Fi, depending on the area of deployment. They use this to communicate with the cloud. But, in rare

cases, if the connection fails, the robots are not stumped. They keep moving safely in permitted routes, till they get reconnected.

During 'rest time' the robots usually move to a hub for charging and software upgrades. This is when diagnostics are uploaded to the cloud. Companies use this to study how the robot's day went, and to improve its behaviour further.

Many technologies, from perception to path planning and mapping, appear common to autonomous delivery robots and autonomous cars. But, Ritukar Vijay, Founder and CEO of Ottonomy.IO explains in one of his articles that, "Although the delivery robots are low in speed, they often engage in more complicated, unstructured scenarios and pedestrian interactions than autonomous cars." Due to this, there are a few areas where autonomous delivery robots are acting as a precursor to autonomous cars.

First, he explains, the nuances of running an autonomous fleet of ten vehicles compared to a fleet of 200 to 2,000 delivery robots is a huge shift in AV operations. The educational benefits of autonomous robot fleets include, and are not limited to, autonomous fleet management, distributed AV information and processing, fallback safety, fleet orches-

tration, and more. Second, because more delivery robots are out in the market, they are more exposed to a variety of edge cases. Sometimes, sidewalks, airports, and curb sides are more challenging than lane-guided roads for autonomous cars, and that results in further advancements in active perception, situational awareness, and autonomous behaviours and safety (in general).

Design, of course, is a different ballgame. The purpose of delivery gives rise to certain special requirements. Obviously, storage is an important aspect of the design. This comes in various forms—modular cargo boxes, simple locker boxes, and so on—depending on the size and purpose of the robot. Some have hot and cold compartments as well. They also need to look friendly and have a safe, harmless design with soft, rounded corners. Some like Nuro even have airbags in the front to protect pedestrians from harm in case of a bump.

Delivery robots have to function in line with local rules and regulations at all times. They can move only where they are permitted. And features like lights and alarms have to be built in, so that fellow pedestrians get a clear indication when and where the robot is moving. Delivery robots also have to be very courteous and nice to people. They are taught to say, *excuse me*, when someone is blocking the way, and *thank you*, after they oblige and move!

"It's a never-ending journey to make sure that this human-robot dynamic is maintained, fostered, and improved. When the robot makes an error or does something unexpected, we try to take all of these points of feedback in to make sure that it gets better and better, because the bar gets raised as well. We can't assume the bar is static. I am of the opinion that if you're going to automate something, it not only needs to be